

Assignment 6

Task-1: (Classes – Data Validation)

Prompt:

Write a simple Python program to create a Student class with name, roll_no, and marks as attributes. Add a method called is_pass() that checks if the student's marks are 40 or above and returns the pass or fail result. Include a constructor and basic validation for the student details. Also create a sample student object and show the output. Keep the code simple and easy to understand

Input:

```
class Student:
```

```
    def __init__(self, name, roll_no, marks):
```

```
        if not name:
```

```
            raise ValueError("Name cannot be empty")
```

```
        if marks < 0 or marks > 100:
```

```
            raise ValueError("Marks should be between 0 and 100")
```

```
        self.name = name
```

```
        self.roll_no = roll_no
```

```
        self.marks = marks
```

```
    def is_pass(self):
```

```
        if self.marks >= 40:
```

```
            return "Pass"
```

```
        else:
```

```
            return "Fail"
```

```
# Create a sample student object
```

```
student1 = Student("Rahul", 101, 75)
```

```
# Display student details
```

```
print("Name:", student1.name)
```

```
print("Roll No:", student1.roll_no)
```

```
print("Marks:", student1.marks)
```

```
print("Result:", student1.is_pass())
```

Assignment 6

output:

```
PS C:\Users\kandi\OneDrive\Desktop\week 5> & C:\Users\kandi\AppData\Local\Python\pythoncore-3.14-64\python.exe "c:/Users/kandi/OneDrive/Desktop/week 5/task-1.py"
Name: Rahul
Roll No: 101
Marks: 75
Result: Pass
```

Assignment 6

Task-2: (Loops – Pattern Generation)

Prompt:

Write a simple Python program with a function to print a right-angled triangle star pattern using a for loop. Then write another function to print the same pattern using a while loop. Keep the code simple, with proper loop conditions, and show the output for both methods.

Input:

Function using for loop

```
def triangle_for(n):
```

```
    for i in range(1, n + 1):
```

```
        print("*" * i)
```

Function using while loop

```
def triangle_while(n):
```

```
    i = 1
```

```
    while i <= n:
```

```
        print("*" * i)
```

```
    i += 1
```

Number of rows

```
n = 5
```

```
print("Using for loop:")
```

```
triangle_for(n)
```

```
print("\nUsing while loop:")
```

```
triangle_while(n)
```

output:

```
PS C:\Users\kandi\OneDrive\Desktop\week 5> & C:\Users\kandi\AppData\Local\Python\pythoncore-3.14-64\python.exe "c:/users/kandi/OneDrive/deskto
p/week 5/task-2.py"
Using for loop:
*
**
***
****
*****

Using while loop:
*
**
***
****
*****
```

Assignment 6

Task-3:(Conditional Statements – Number Analysis)

Prompt:

Write a simple Python function that takes a number as input and checks whether it is positive, negative, or zero using if-elif-else statements. Test the function with multiple different numbers and display the result for each input. Keep the code simple and easy to understand.

Input:

```
def check_number(num):  
    if num > 0:  
        return "Positive"  
    elif num < 0:  
        return "Negative"  
    else:  
        return "Zero"  
  
# Test the function with different numbers  
numbers = [10, -5, 0, 25, -12]  
  
for num in numbers:  
    print(num, "is", check_number(num))
```

output:

```
PS C:\Users\kandi\OneDrive\Desktop\week 5> & C:\Users\kandi\AppData\Local\Python\pythoncore-3.14-64\python.exe "c:/Users/kandi/OneDrive/Desktop/week 5/task-3.py"  
10 is Positive  
-5 is Negative  
0 is Zero  
25 is Positive  
-12 is Negative
```

Assignment 6

Task-4: (Nested Conditionals)

Prompt:

Write a simple Python function called `check_discount(age, is_member)` to determine a person's discount eligibility. Use nested if-else statements where people aged 60 or above get a senior discount, and members get an additional discount. Test the function with a few different age and membership combinations and display the results. Keep the code simple and easy to understand.

Input:

```
def check_discount(age, is_member):  
    if age >= 60:  
        if is_member:  
            return "Senior discount + Member discount"  
        else:  
            return "Senior discount"  
    else:  
        if is_member:  
            return "Member discount"  
        else:  
            return "No discount"  
  
# Test the function  
print("Age: 65, Member: True ->", check_discount(65, True))  
print("Age: 65, Member: False ->", check_discount(65, False))  
print("Age: 30, Member: True ->", check_discount(30, True))  
print("Age: 30, Member: False ->", check_discount(30, False))
```

output:

```
PS C:\Users\kandi\OneDrive\Desktop\week 5> & C:\Users\kandi\AppData\Local\python\pythoncore-3.14-64\python.exe "c:/Users/kandi/OneDrive/Desktop/week 5/task-4.py"  
Age: 65, Member: True -> Senior discount + Member discount  
Age: 65, Member: False -> Senior discount  
Age: 30, Member: True -> Member discount  
Age: 30, Member: False -> No discount
```

Assignment 6

Task-5: (Class – Mathematical Opera)

Prompt:

Write a simple Python program to create a Circle class that takes the radius as input. Add methods called `area()` and `circumference()` to calculate the area and circumference of the circle. Create an object with a sample radius, display the results, and briefly explain the code. Keep the program simple and easy to understand.

Input:

```
import math

class Circle:

    def __init__(self, radius):

        self.radius = radius

    def area(self):

        return math.pi * self.radius * self.radius

    def circumference(self):

        return 2 * math.pi * self.radius

# Create a Circle object

circle1 = Circle(5)

# Display the results

print("Radius:", circle1.radius)

print("Area:", circle1.area())

print("Circumference:", circle1.circumference())
```

output:

```
PS C:\Users\kandi\OneDrive\Desktop\week 5> & C:\Users\kandi\AppData\Local\Python\pythoncore-3.14-64\python.exe "c:/Users/kandi/OneDrive/Desktop/week 5/task-5.py"
Radius: 5
Area: 78.53981633974483
Circumference: 31.41592653589793
```